共同設定都是夏月、平日、晴天

S1 測試 2 段式 無累計電價 無停電 無設備啟用

📩 收到的 config： {'pricing': 'two_stage', 'season': 'summer', 'day_type': 'weekday', 'weather': 'sunny', 'monthly_cumulative_kwh': 0, 'outage': None, 'device_schedules': []}

🔁 [重試檢查] 第 0 次嘗試，目前 bess_override=None

🔎 [天氣檢查] PSO 這次使用的 PV 檔案：C:\projects\hems-simulation-web\data\01_raw\pv_curve_15min.csv

🔎 [PSO 輸入比對] 13:00~15:00 PV: [4.78867, 4.6735299999999995, 4.56147, 4.415, 4.25413, 4.07467, 3.8710700000000005, 3.6598]

🔎 [PSO 輸入比對] 13:00~15:00 critical_load_kw: [0.10398015000000001, 0.104320565, 0.105352634, 0.10483838199999998, 0.105803373, 0.10691419400000002, 0.10865781599999999, 0.11004641800000001]

🎯 [PSO 自評] PSO 演算法自己算出來的理論總成本(電費+懲罰項): -162.48927667849046

🔎 [天氣檢查] 離線驗證這次使用的 PV 檔案：pv_curve_15min.csv

🔍 [除錯] 目前實際使用的電池指令是: New Storage.Battery_Sys phases=1 bus1=Inv_AC.1.2 kV=0.22 kVA=6.0 kWrated=5.0 kWhrated=20.0 pf=1.0 %stored=40 %reserve=0 %IdlingKw=0 %EffCharge=95 %EffDischarge=95 DispMode=External

基準電路建置完成！開始進行 15 分鐘步進模擬 (模式: PSO)..

F12 Response

1. {status: "success", message: "切換成功", pso_total_cost: 24.71, baseline_total_cost: 2.65,…}
   - baseline_adjusted_cost: -101.17
   - baseline_battery_asset_value: 0
   - baseline_export_kwh: 18.447
   - baseline_export_revenue: 103.82
   - baseline_had_overload: false
   - baseline_total_cost: 2.65
   - message: "切換成功"
   - pso_adjusted_cost: -161.77
   - pso_battery_asset_value: 0
   - pso_export_kwh: 33.135
   - pso_export_revenue: 186.48
   - pso_total_cost: 24.71
   - savings: 60.6
   - status: "success"

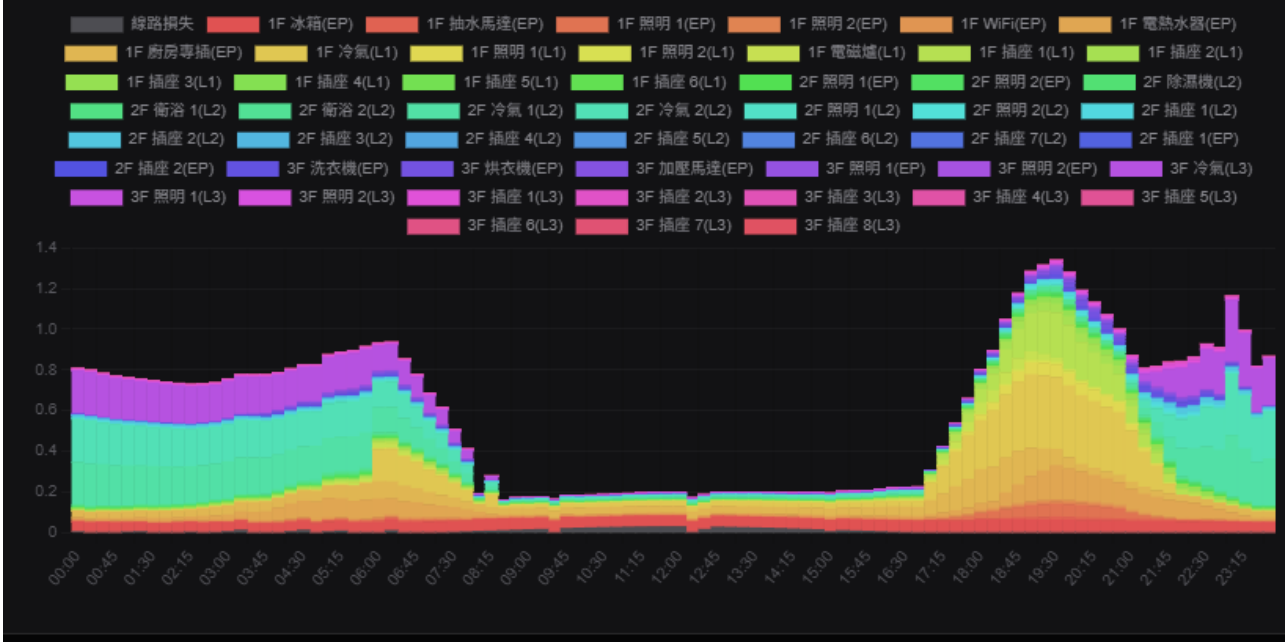
今日總用電量

14.3 度

啟用設備數

35 台

HOURLY LOAD (KW)



優化前電價

NT$ -101.2

優化後電價

NT$ -161.8

省下金額

NT$ 60.6

02 — OPTIMIZATION

POS 儲能排程最佳化

於用電低谷引導放電、太陽能充裕時段安排充電，降低尖峰購電功率並平滑全日負載。

1. GRID PURCHASE POWER (KW) - 電網購電

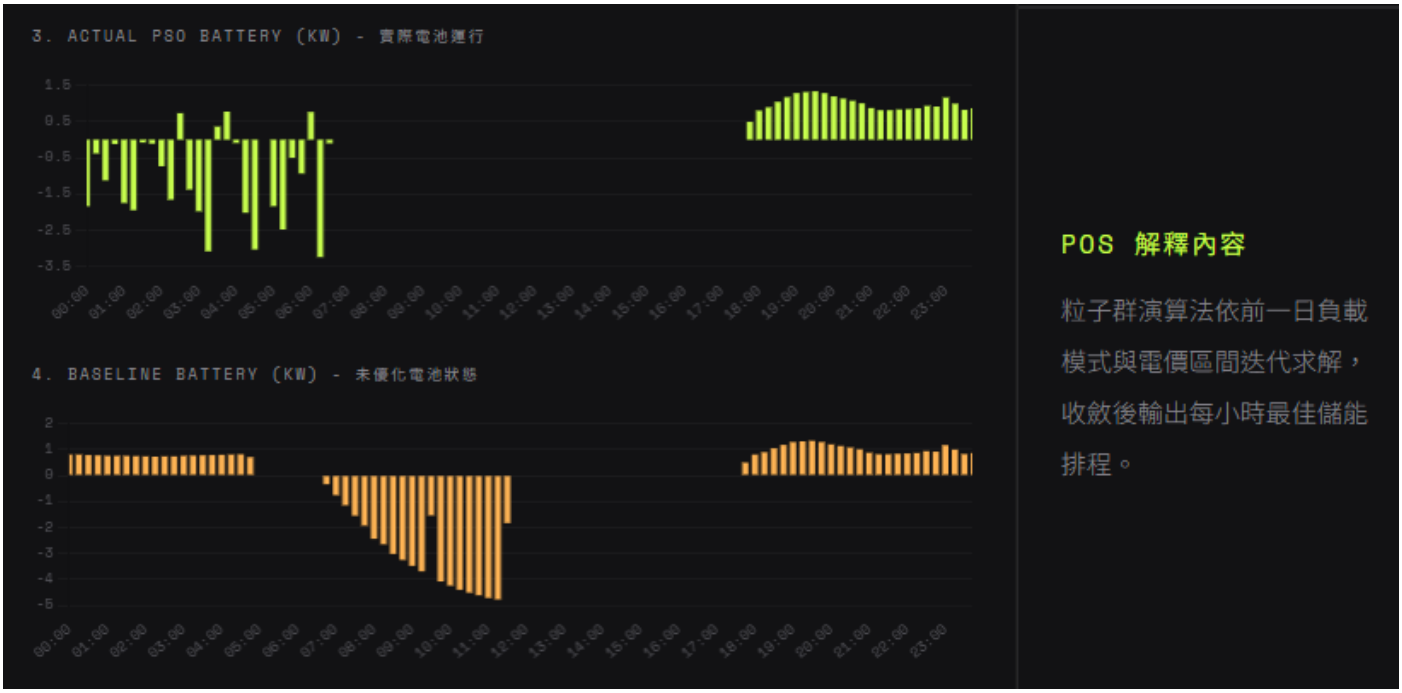


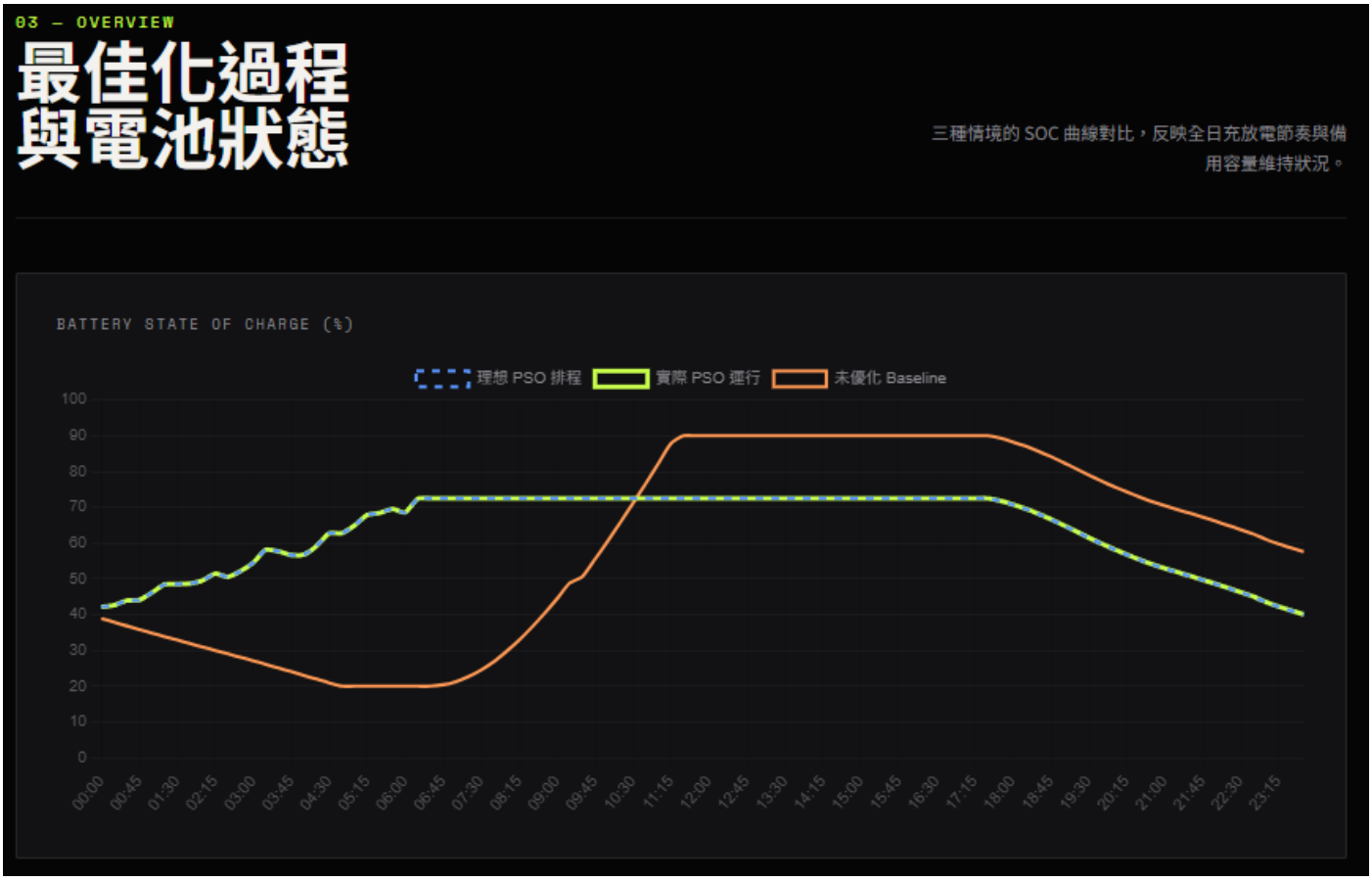
2. IDEAL PSO BATTERY (KW) - 理想電池排程



數據解析

系統逐時段求解電池最佳充放電策略，在維持用電舒適度的前提下最小化尖峰負載與總電費支出。





pso 起始與結束電量

Time	soc
00:00	42.17
23:45	40.05

Baseline 的起始與結束電量

Time	soc
00:00	38.95
23:45	57.55

S2 測試 3 段式 無累計電價 無停電 無設備啟用

📩 收到的 config： {'pricing': 'three_stage', 'season': 'summer', 'day_type': 'weekday', 'weather': 'sunny', 'monthly_cumulative_kwh': 0, 'outage': None, 'device_schedules': []}

🔁 [重試檢查] 第 0 次嘗試，目前 bess_override=None

🔎 [天氣檢查] PSO 這次使用的 PV 檔案：C:\projects\hems-simulation-web\data\01_raw\pv_curve_15min.csv

🔎 [PSO 輸入比對] 13:00~15:00 PV: [4.78867, 4.6735299999999995, 4.56147, 4.415, 4.25413, 4.07467, 3.8710700000000005, 3.6598]

🔎 [PSO 輸入比對] 13:00~15:00 critical_load_kw: [0.10398015000000001, 0.104320565, 0.105352634, 0.10483838199999998, 0.105803373, 0.10691419400000002, 0.10865781599999999, 0.11004641800000001]

🎯 [PSO 自評] PSO 演算法自己算出來的理論總成本(電費+懲罰項): -155.0460113844004

🔎 [天氣檢查] 離線驗證這次使用的 PV 檔案：pv_curve_15min.csv

🔍 [除錯] 目前實際使用的電池指令是: New Storage.Battery_Sys phases=1 bus1=Inv_AC.1.2 kV=0.22 kVA=6.0 kWrated=5.0 kWhrated=20.0 pf=1.0 %stored=40 %reserve=0 %IdlingKw=0 %EffCharge=95 %EffDischarge=95 DispMode=External

基準電路建置完成！開始進行 15 分鐘步進模擬 (模式: PSO)...

F12 Response

1.

{status: "success", message: "切換成功", pso_total_cost: 32.2, baseline_total_cost: 2.67,…}

baseline_adjusted_cost: -101.15

baseline_battery_asset_value: 0

baseline_export_kwh: 18.447

baseline_export_revenue: 103.82

baseline_had_overload: false

baseline_total_cost: 2.67

message: "切換成功"

pso_adjusted_cost: -154.28

pso_battery_asset_value: 0

pso_export_kwh: 33.135

pso_export_revenue: 186.48

pso_total_cost: 32.2

savings: 53.13

status: "success"

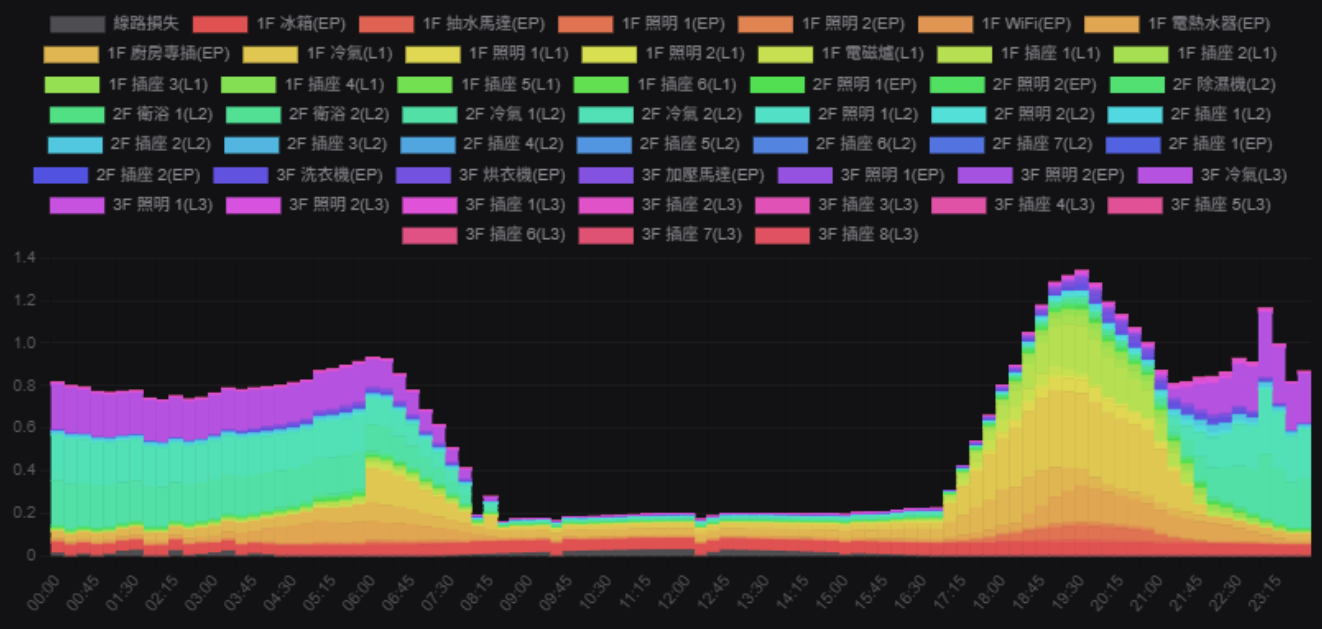
今日總用電量

14.3 度

啟用設備數

35 台

HOURLY LOAD (KW)



優化前電價

NT$ -101.2

優化後電價

NT$ -154.3

省下金額

NT$ 53.1

02 — OPTIMIZATION

POS 儲能排程最佳化

於用電低谷引導放電、太陽能充裕時段安排充電，降低尖峰購電功率並平滑全日負載。

1. GRID PURCHASE POWER (KW) - 電網購電



2. IDEAL PSO BATTERY (KW) - 理想電池排程



數據解析

系統逐時段求解電池最佳充放電策略，在維持用電舒適度的前提下最小化尖峰負載與總電費支出。

3. ACTUAL PSO BATTERY (KW) - 實際電池運行

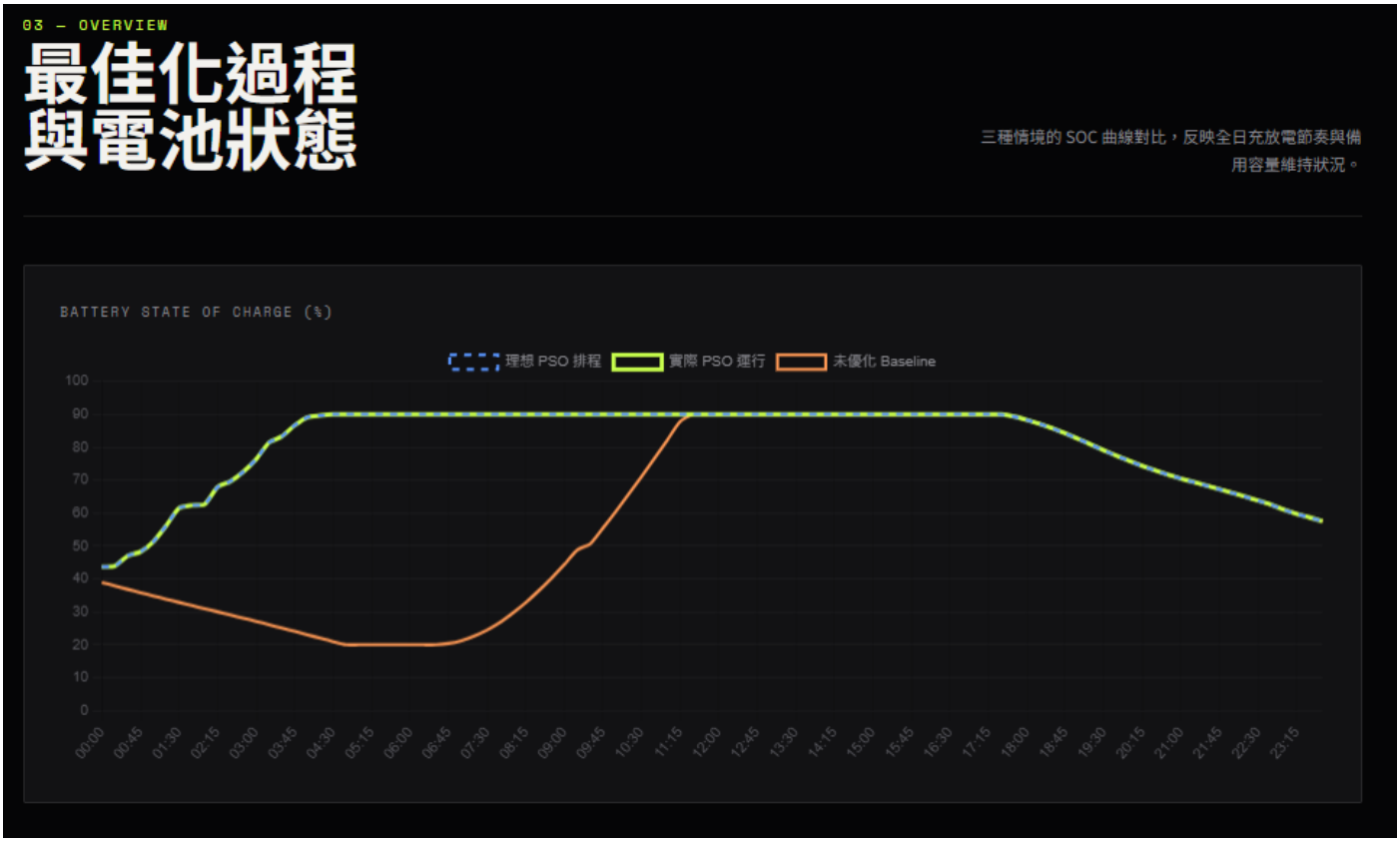


4. BASELINE BATTERY (KW) - 未優化電池狀態



POS 解釋內容

粒子群演算法依前一日負載模式與電價區間迭代求解，收斂後輸出每小時最佳儲能排程。



pso 起始與結束電量

Time	soc
00:00	43.69
23:45	57.55

Baseline 的起始與結束電量

Time	soc
00:00	38.95
23:45	57.55

S3 測試 累計電價 已累計 0 無停電 無設備啟用

📩 收到的 config： {'pricing': 'progressive', 'season': 'summer', 'day_type': 'weekday', 'weather': 'sunny', 'monthly_cumulative_kwh': 0, 'outage': None, 'device_schedules': []}

🔁 [重試檢查] 第 0 次嘗試，目前 bess_override=None

🔎 [天氣檢查] PSO 這次使用的 PV 檔案：C:\projects\hems-simulation-web\data\01_raw\pv_curve_15min.csv

🔎 [PSO 輸入比對] 13:00~15:00 PV: [4.78867, 4.6735299999999995, 4.56147, 4.415, 4.25413, 4.07467, 3.8710700000000005, 3.6598]

🔎 [PSO 輸入比對] 13:00~15:00 critical_load_kw: [0.10398015000000001, 0.104320565, 0.105352634, 0.10483838199999998, 0.105803373, 0.10691419400000002, 0.10865781599999999, 0.11004641800000001]

🎯 [PSO 自評] PSO 演算法自己算出來的理論總成本(電費+懲罰項): -167.14736597677623

🔎 [天氣檢查] 離線驗證這次使用的 PV 檔案：pv_curve_15min.csv

🔍 [除錯] 目前實際使用的電池指令是: New Storage.Battery_Sys phases=1 bus1=Inv_AC.1.2 kV=0.22 kVA=6.0 kWrated=5.0 kWhrated=20.0 pf=1.0 %stored=40 %reserve=0 %IdlingKw=0 %EffCharge=95 %EffDischarge=95 DispMode=External

基準電路建置完成！開始進行 15 分鐘步進模擬 (模式: PSO)...

F12 Response

1.

{status: "success", message: "切換成功", pso_total_cost: 19.96, baseline_total_cost: 2.24,…}

baseline_adjusted_cost: -101.58

baseline_battery_asset_value: 0

baseline_export_kwh: 18.447

baseline_export_revenue: 103.82

baseline_had_overload: false

baseline_total_cost: 2.24

message: "切換成功"

pso_adjusted_cost: -166.52

pso_battery_asset_value: 0

pso_export_kwh: 33.135

pso_export_revenue: 186.48

pso_total_cost: 19.96

savings: 64.94

status: "success"

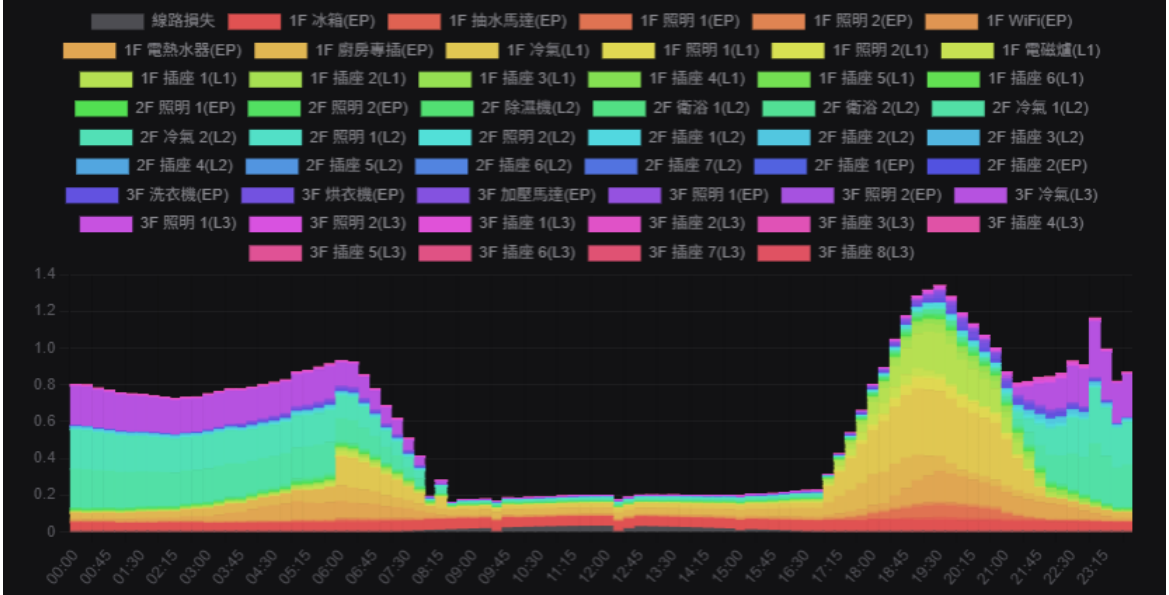
今日總用電量

14.3 度

啟用設備數

35 台

HOURLY LOAD (KW)



優化前電價

NT$ -101.6

優化後電價

NT$ -166.5

省下金額

NT$ 64.9

02 — OPTIMIZATION

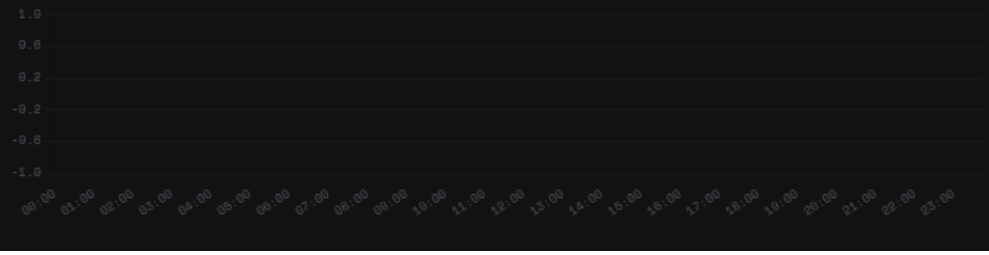
POS 儲能排程最佳化

於用電低谷引導放電、太陽能充裕時段安排充電，降低尖峰購電功率並平滑全日負載。

1. GRID PURCHASE POWER (KW) - 電網購電

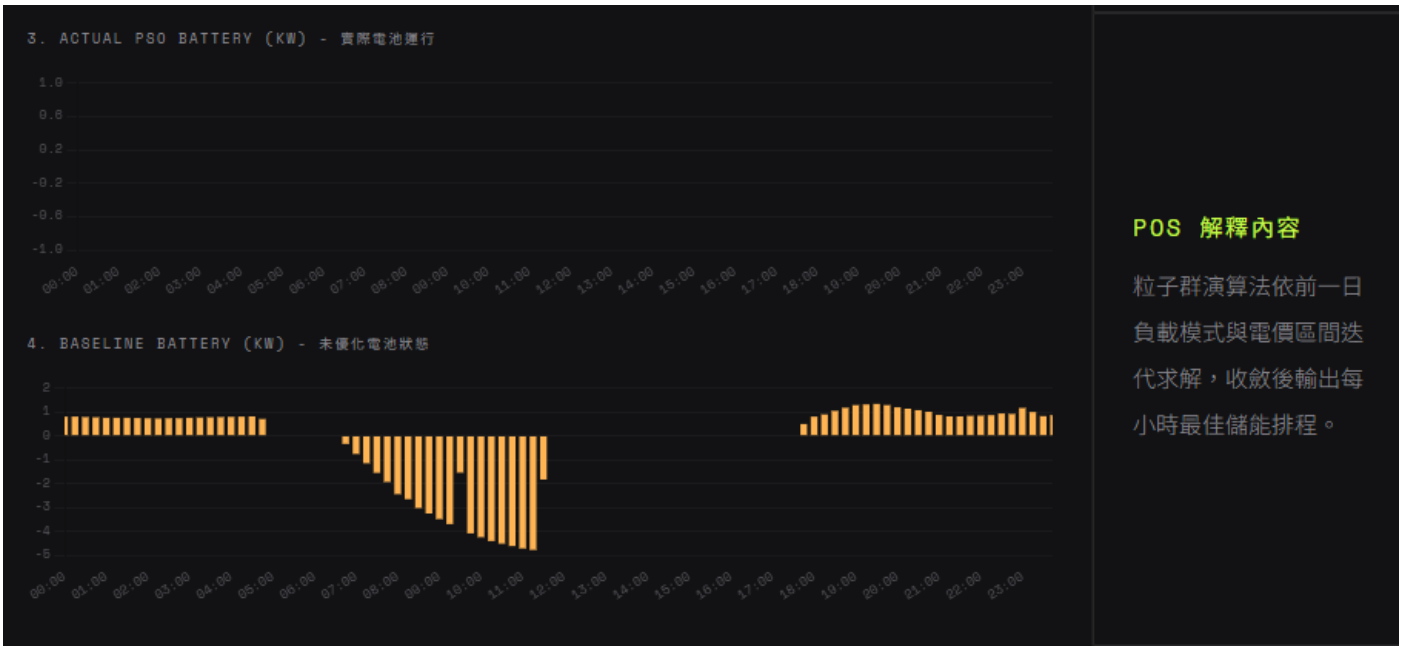


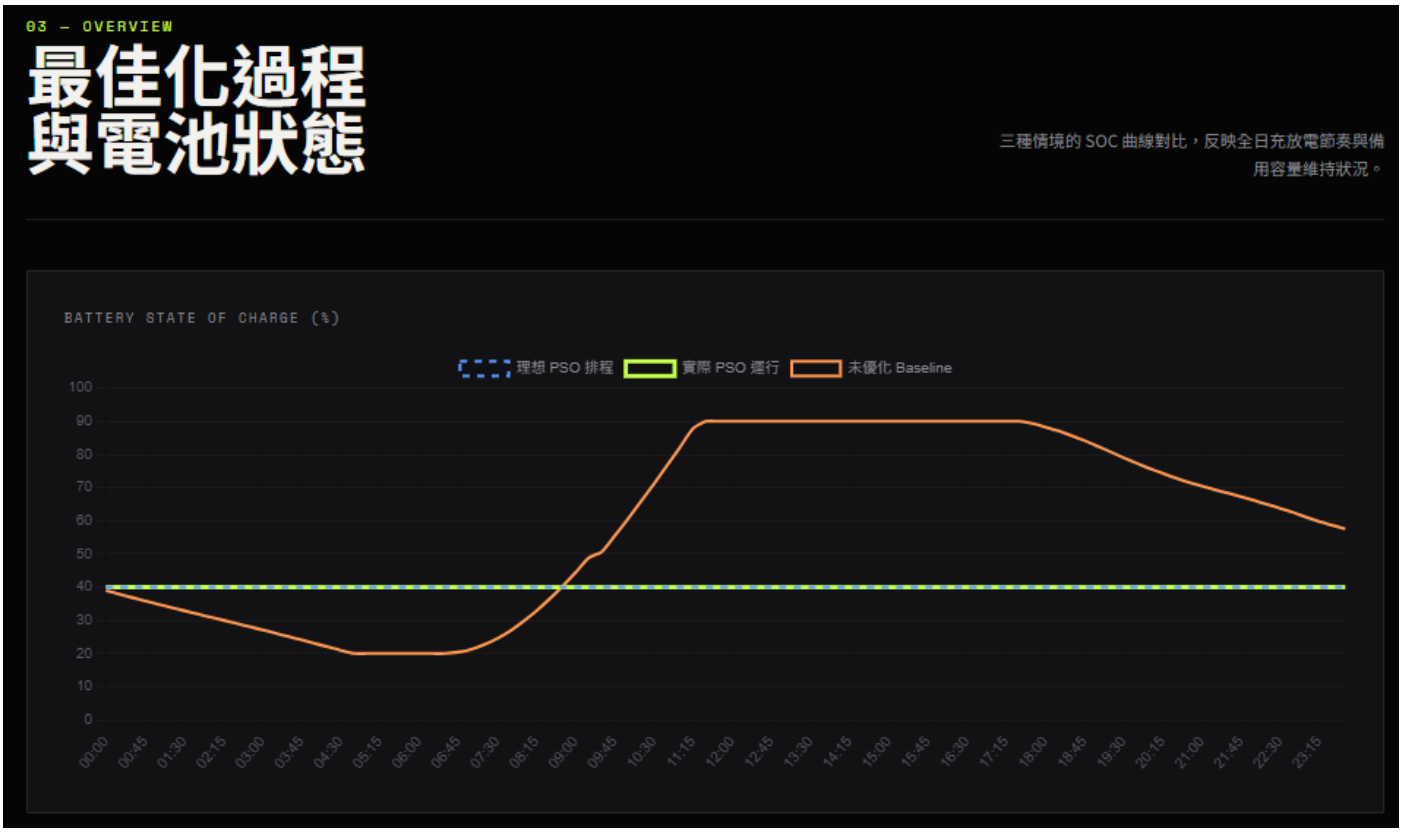
2. IDEAL PSO BATTERY (KW) - 理想電池排程



數據解析

系統逐時段求解電池最佳充放電策略，在維持用電舒適度的前提下最小化尖峰負載與總電費支出。





pso 起始與結束電量

Time	soc
00:00	40
23:45	40

Baseline 的起始與結束電量

Time	soc
00:00	38.95
23:45	57.55

S4 測試 累計電價 已累計 800 無設備啟用

📥 收到的 config： {'pricing': 'progressive', 'season': 'summer', 'day_type': 'weekday', 'weather': 'sunny', 'monthly_cumulative_kwh': 800, 'outage': None, 'device_schedules': []}

🔁 [重試檢查] 第 0 次嘗試，目前 bess_override=None

🔎 [天氣檢查] PSO 這次使用的 PV 檔案：C:\projects\hems-simulation-web\data\01_raw\pv_curve_15min.csv

🔎 [PSO 輸入比對] 13:00~15:00 PV: [4.78867, 4.6735299999999995, 4.56147, 4.415, 4.25413, 4.07467, 3.8710700000000005, 3.6598]

🔎 [PSO 輸入比對] 13:00~15:00 critical_load_kw: [0.10398015000000001, 0.104320565, 0.105352634, 0.10483838199999998, 0.105803373, 0.10691419400000002, 0.10865781599999999, 0.11004641800000001]

🎯 [PSO 自評] PSO 演算法自己算出來的理論總成本(電費+懲罰項): -116.80305371592335

🔎 [天氣檢查] 離線驗證這次使用的 PV 檔案：pv_curve_15min.csv

🔍 [除錯] 目前實際使用的電池指令是: New Storage.Battery_Sys phases=1 bus1=Inv_AC.1.2 kV=0.22 kVA=6.0 kWrated=5.0 kWhrated=20.0 pf=1.0 %stored=40 %reserve=0 %IdlingKw=0 %EffCharge=95 %EffDischarge=95 DispMode=External

基準電路建置完成！開始進行 15 分鐘步進模擬 (模式: PSO)...

F12 Response

1. {status: "success", message: "切換成功", pso_total_cost: 12.61, baseline_total_cost: 8.09,…}
   - baseline_adjusted_cost: -95.73
   - baseline_battery_asset_value: 0
   - baseline_export_kwh: 18.447
   - baseline_export_revenue: 103.82
   - baseline_had_overload: false
   - baseline_total_cost: 8.09
   - message: "切換成功"
   - pso_adjusted_cost: -116.28
   - pso_battery_asset_value: 0
   - pso_export_kwh: 22.901
   - pso_export_revenue: 128.89
   - pso_total_cost: 12.61
   - savings: 20.55
   - status: "success"

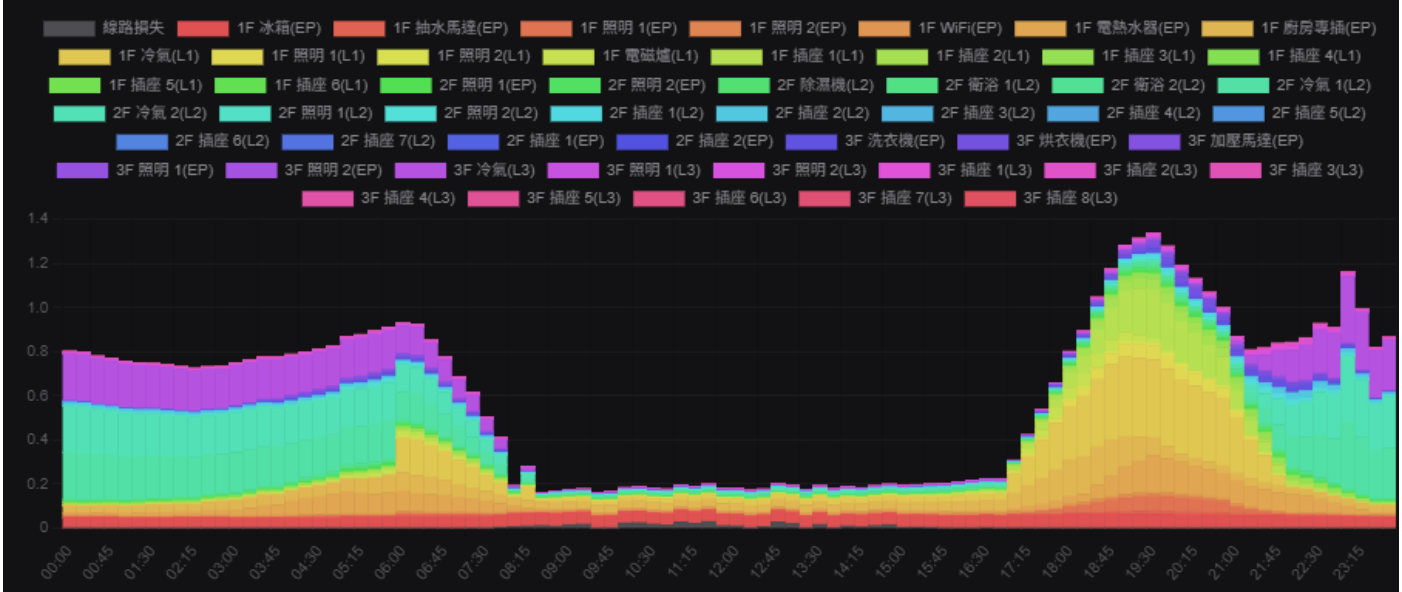
今日總用電量

14.3 度

啟用設備數

35 台

HOURLY LOAD (KW)



優化前電價

NT$ -95.7

優化後電價

NT$ -116.3

省下金額

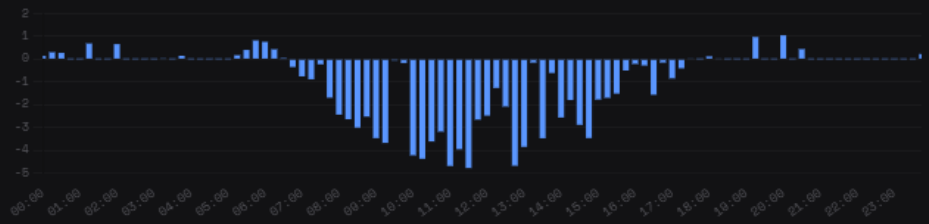
NT$ 20.5

02 — OPTIMIZATION

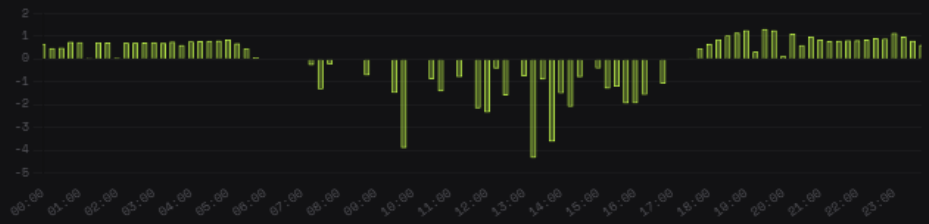
POS 儲能排程最佳化

於用電低谷引導放電、太陽能充裕時段安排充電，降低尖峰購電功率並平滑全日負載。

1. GRID PURCHASE POWER (KW) - 電網購電



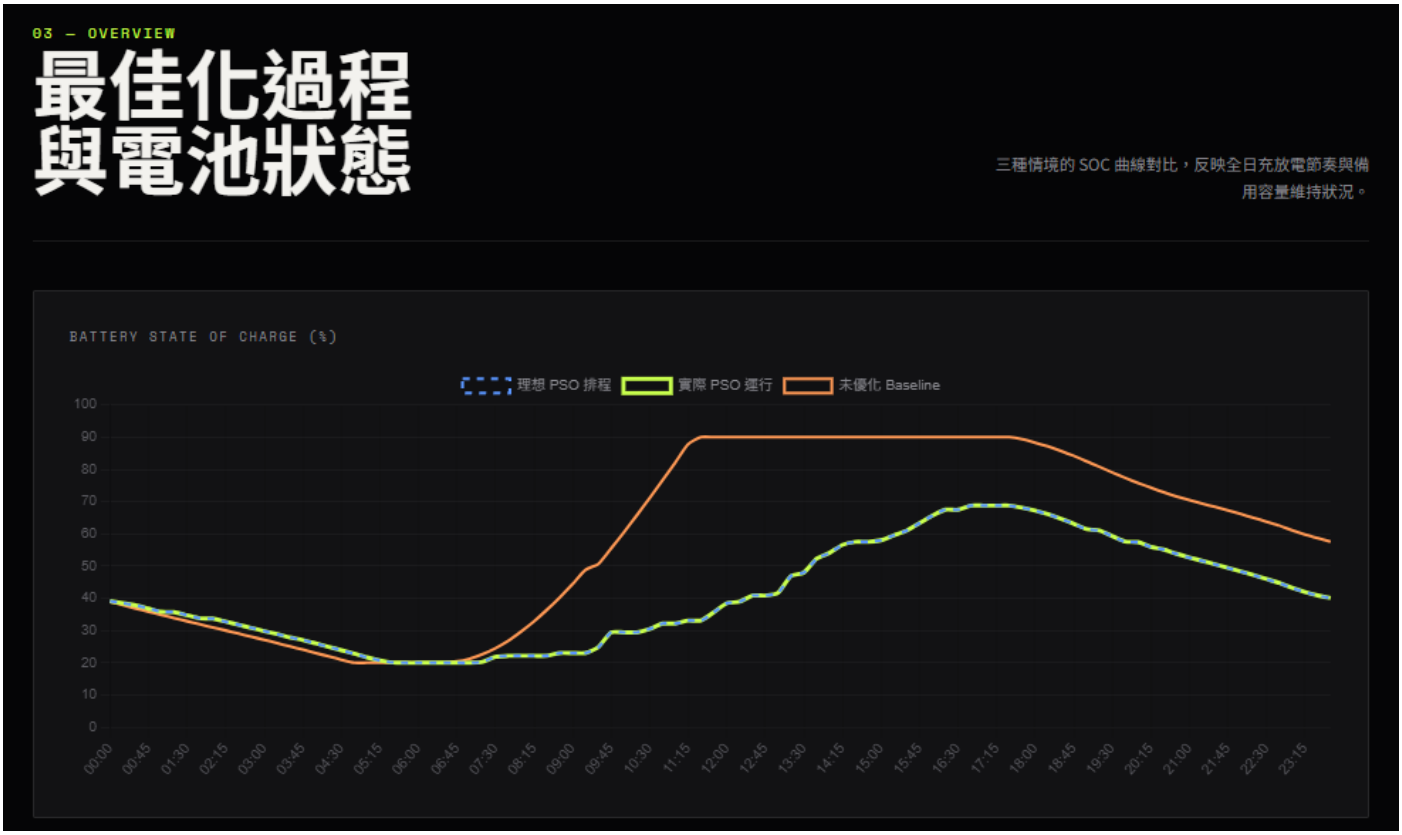
2. IDEAL PSO BATTERY (KW) - 理想電池排程



數據解析

系統逐時段求解電池最佳充放電策略，在維持用電舒適度的前提下最小化尖峰負載與總電費支出。





pso 起始與結束電量

Time	soc
00:00	39.13
23:45	40.06

Baseline 的起始與結束電量

Time	soc
00:00	38.95
23:45	57.55

S5 測試 2 段式 無累計電價 有停電 18:00~21:00 無設備啟用

📩 收到的 config： {'pricing': 'two_stage', 'season': 'summer', 'day_type': 'weekday', 'weather': 'sunny', 'monthly_cumulative_kwh': 0, 'outage': {'start_time': '18:00', 'end_time': '21:00'}, 'device_schedules': []}

🔁 [重試檢查] 第 0 次嘗試，目前 bess_override=None

🔎 [天氣檢查] PSO 這次使用的 PV 檔案：C:\projects\hems-simulation-web\data\01_raw\pv_curve_15min.csv

🔎 [PSO 輸入比對] 13:00~15:00 PV: [4.78867, 4.6735299999999995, 4.56147, 4.415, 4.25413, 4.07467, 3.8710700000000005, 3.6598]

🔎 [PSO 輸入比對] 13:00~15:00 critical_load_kw: [0.10398015000000001, 0.104320565, 0.105352634, 0.10483838199999998, 0.105803373, 0.10691419400000002, 0.10865781599999999, 0.11004641800000001]

🎯 [PSO 自評] PSO 演算法自己算出來的理論總成本(電費+懲罰項): -167.2954456978664

🔎 [天氣檢查] 離線驗證這次使用的 PV 檔案：pv_curve_15min.csv

🔍 [除錯] 目前實際使用的電池指令是: New Storage.Battery_Sys phases=1 bus1=Inv_AC.1.2 kV=0.22 kVA=6.0 kWrated=5.0 kWhrated=20.0 pf=1.0 %stored=40 %reserve=0 %IdlingKw=0 %EffCharge=95 %EffDischarge=95 DispMode=External

基準電路建置完成！開始進行 15 分鐘步進模擬 (模式: PSO)...

F12 Response

1. {status: "success", message: "切換成功", pso_total_cost: 19.89, baseline_total_cost: 2.64,…}
   - baseline_adjusted_cost: -101.18
   - baseline_battery_asset_value: 0
   - baseline_export_kwh: 18.447
   - baseline_export_revenue: 103.82
   - baseline_had_overload: false
   - baseline_total_cost: 2.64
   - message: "切換成功"
   - pso_adjusted_cost: -166.59
   - pso_battery_asset_value: 0
   - pso_export_kwh: 33.135
   - pso_export_revenue: 186.48
   - pso_total_cost: 19.89
   - savings: 65.41
   - status: "success"

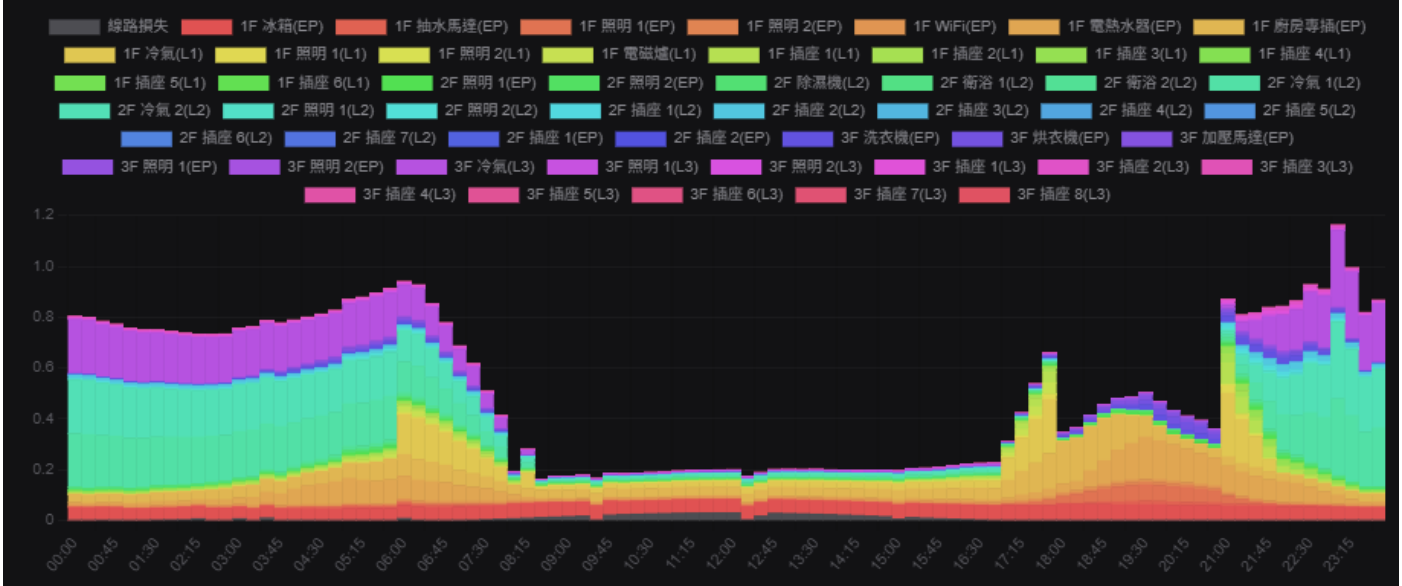
今日總用電量

12.2 度

啟用設備數

35 台

HOURLY LOAD (KW)



優化前電價

NT$ -101.2

優化後電價

NT$ -166.6

省下金額

NT$ 65.4

02 — OPTIMIZATION

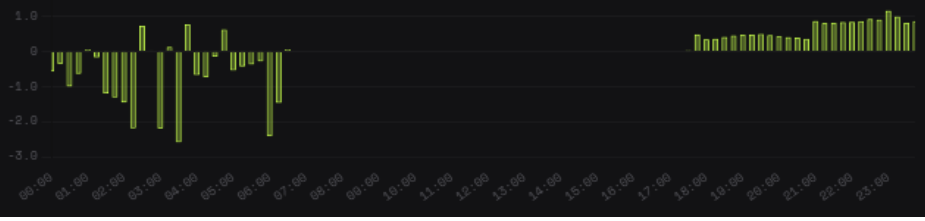
POS 儲能排程最佳化

於用電低谷引導放電、太陽能充裕時段安排充電，降低尖峰購電功率並平滑全日負載。

1. GRID PURCHASE POWER (KW) - 電網購電

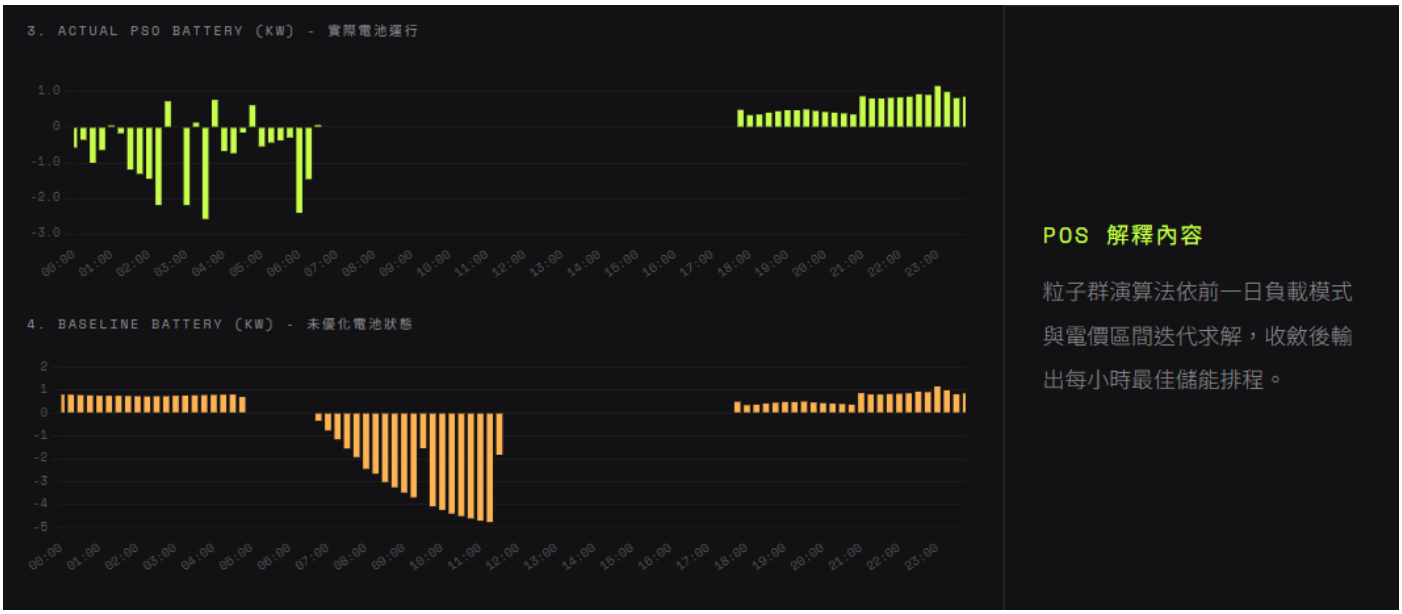


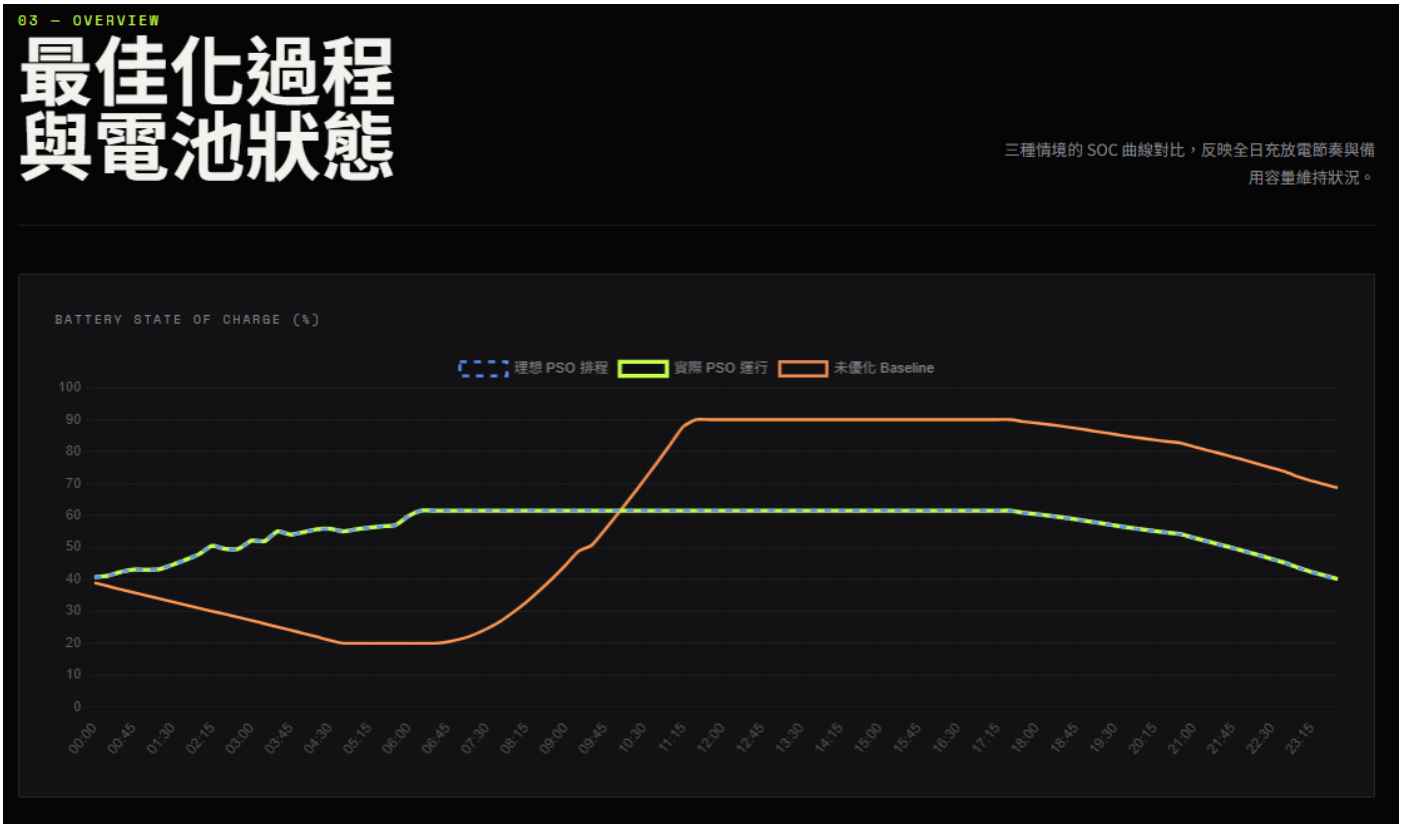
2. IDEAL PSO BATTERY (KW) - 理想電池排程



數據解析

系統逐時段求解電池最佳充放電策略，在維持用電舒適度的前提下最小化尖峰負載與總電費支出。





Pso 的電量紀錄

Time	soc	battery_power_kw	current_a
00:00	40.68	-0.57	2.62
00:15	41.1	-0.35	1.61
00:30	42.29	-1	4.56
00:45	43.05	-0.64	2.91
01:00	42.97	0.05	0.25
01:15	43.18	-0.17	0.79
01:30	44.6	-1.19	5.45
01:45	46.15	-1.31	5.98
02:00	47.88	-1.45	6.64

02:15	50.47	-2.19	10
02:30	49.51	0.73	3.33
02:45	49.51	0	0
03:00	52.11	-2.19	10.01
03:15	51.94	0.13	0.59
03:30	55.01	-2.58	11.81
03:45	53.99	0.77	3.53
04:00	54.79	-0.67	3.07
04:15	55.66	-0.73	3.32
04:30	55.84	-0.15	0.7
04:45	55.02	0.62	2.85
05:00	55.66	-0.54	2.48
05:15	56.17	-0.43	1.96
05:30	56.61	-0.37	1.7
05:45	56.95	-0.29	1.3
06:00	59.81	-2.41	11.01
06:15	61.55	-1.46	6.68
06:30	61.48	0.06	0.26
06:45	61.48	0	0
07:00	61.48	0	0
07:15	61.48	0	0
07:30	61.48	0	0
07:45	61.48	0	0
08:00	61.48	0	0
08:15	61.48	0	0
08:30	61.48	0	0
08:45	61.48	0	0
09:00	61.48	0	0
09:15	61.48	0	0
09:30	61.48	0	0
09:45	61.48	0	0
10:00	61.48	0	0
10:15	61.48	0	0
10:30	61.48	0	0
10:45	61.48	0	0
11:00	61.48	0	0
11:15	61.48	0	0
11:30	61.48	0	0
11:45	61.48	0	0
12:00	61.48	0	0
12:15	61.48	0	0
12:30	61.48	0	0

12:45	61.48	0	0
13:00	61.48	0	0
13:15	61.48	0	0
13:30	61.48	0	0
13:45	61.48	0	0
14:00	61.48	0	0
14:15	61.48	0	0
14:30	61.48	0	0
14:45	61.48	0	0
15:00	61.48	0	0
15:15	61.48	0	0
15:30	61.48	0	0
15:45	61.48	0	0
16:00	61.48	0	0
16:15	61.48	0	0
16:30	61.48	0	0
16:45	61.48	0	0
17:00	61.48	0	0
17:15	61.48	0	0
17:30	61.48	0	0
17:45	60.84	0.49	2.21
18:00	60.38	0.34	1.56
18:15	59.91	0.36	1.65
18:30	59.36	0.41	1.88
18:45	58.77	0.45	2.06
19:00	58.13	0.48	2.18
19:15	57.5	0.48	2.2
19:30	56.84	0.5	2.27
19:45	56.23	0.46	2.11
20:00	55.66	0.43	1.96
20:15	55.12	0.41	1.86
20:30	54.61	0.39	1.78
20:45	54.14	0.36	1.63
21:00	53	0.87	3.95
21:15	51.94	0.81	3.67
21:30	50.86	0.81	3.71
21:45	49.77	0.83	3.8
22:00	48.66	0.84	3.83
22:15	47.53	0.86	3.92
22:30	46.31	0.93	4.22
22:45	45.12	0.91	4.13
23:00	43.59	1.16	5.28

23:15	42.29	0.99	4.51
23:30	41.21	0.82	3.72
23:45	40.07	0.86	3.94

Baseline 的電池紀錄

Time	soc	battery_power_kw	current_a
00:00	38.95	0.8	3.65
00:15	37.9	0.8	3.63
00:30	36.87	0.78	3.55
00:45	35.86	0.77	3.5
01:00	34.87	0.75	3.44
01:15	33.88	0.75	3.41
01:30	32.9	0.75	3.4
01:45	31.93	0.74	3.36
02:00	30.97	0.73	3.33
02:15	30.02	0.72	3.29
02:30	29.06	0.73	3.33
02:45	28.09	0.73	3.33
03:00	27.11	0.75	3.41
03:15	26.11	0.76	3.47
03:30	25.09	0.77	3.53
03:45	24.07	0.78	3.53
04:00	23.04	0.79	3.58
04:15	21.99	0.8	3.63
04:30	20.92	0.81	3.69
04:45	20	0.7	3.19
05:00	20	0	0
05:15	20	0	0
05:30	20	0	0
05:45	20	0	0
06:00	20	0	0
06:15	20	0	0
06:30	20	0	0
06:45	20.41	-0.34	1.56
07:00	21.3	-0.76	3.44
07:15	22.67	-1.15	5.24
07:30	24.51	-1.55	7.07
07:45	26.8	-1.93	8.79
08:00	29.7	-2.44	11.14
08:15	32.85	-2.65	12.06
08:30	36.43	-3.02	13.75
08:45	40.28	-3.25	14.79

09:00	44.42	-3.48	15.86
09:15	48.81	-3.69	16.84
09:30	50.64	-1.54	7.03
09:45	55.48	-4.08	18.59
10:00	60.52	-4.24	19.34
10:15	65.74	-4.4	20.04
10:30	71.1	-4.51	20.55
10:45	76.57	-4.61	21.02
11:00	82.17	-4.71	21.48
11:15	87.83	-4.77	21.73
11:30	90	-1.83	8.29
11:45	90	0	0
12:00	90	0	0
12:15	90	0	0
12:30	90	0	0
12:45	90	0	0
13:00	90	0	0
13:15	90	0	0
13:30	90	0	0
13:45	90	0	0
14:00	90	0	0
14:15	90	0	0
14:30	90	0	0
14:45	90	0	0
15:00	90	0	0
15:15	90	0	0
15:30	90	0	0
15:45	90	0	0
16:00	90	0	0
16:15	90	0	0
16:30	90	0	0
16:45	90	0	0
17:00	90	0	0
17:15	90	0	0
17:30	90	0	0
17:45	89.36	0.49	2.21
18:00	88.91	0.34	1.56
18:15	88.43	0.36	1.65
18:30	87.89	0.41	1.88
18:45	87.29	0.45	2.06
19:00	86.66	0.48	2.18
19:15	86.02	0.48	2.2

19:30	85.37	0.5	2.27
19:45	84.75	0.46	2.11
20:00	84.19	0.43	1.96
20:15	83.65	0.41	1.86
20:30	83.13	0.39	1.78
20:45	82.66	0.36	1.63
21:00	81.52	0.87	3.95
21:15	80.46	0.81	3.67
21:30	79.39	0.81	3.71
21:45	78.29	0.83	3.8
22:00	77.18	0.84	3.83
22:15	76.05	0.86	3.92
22:30	74.83	0.93	4.22
22:45	73.64	0.91	4.13
23:00	72.12	1.16	5.28
23:15	70.81	0.99	4.51
23:30	69.74	0.82	3.72
23:45	68.6	0.86	3.94

S6 測試 2 段式 無累計電價 有停電 21:00~23:00 無設備啟用

📩 收到的 config： {'pricing': 'two_stage', 'season': 'summer', 'day_type': 'weekday', 'weather': 'sunny', 'monthly_cumulative_kwh': 0, 'outage': {'start_time': '21:00', 'end_time': '23:00'}, 'device_schedules': []}

🔁 [重試檢查] 第 0 次嘗試，目前 bess_override=None

🔎 [天氣檢查] PSO 這次使用的 PV 檔案：C:\projects\hems-simulation-web\data\01_raw\pv_curve_15min.csv

🔎 [PSO 輸入比對] 13:00~15:00 PV: [4.78867, 4.6735299999999995, 4.56147, 4.415, 4.25413, 4.07467, 3.8710700000000005, 3.6598]

🔎 [PSO 輸入比對] 13:00~15:00 critical_load_kw: [0.10398015000000001, 0.104320565, 0.105352634, 0.10483838199999998, 0.105803373, 0.10691419400000002, 0.10865781599999999, 0.11004641800000001]

🎯 [PSO 自評] PSO 演算法自己算出來的理論總成本(電費+懲罰項): -165.18417184463652

🔎 [天氣檢查] 離線驗證這次使用的 PV 檔案：pv_curve_15min.csv

🔍 [除錯] 目前實際使用的電池指令是: New Storage.Battery_Sys phases=1 bus1=Inv_AC.1.2 kV=0.22 kVA=6.0 kWrated=5.0 kWhrated=20.0 pf=1.0 %stored=40 %reserve=0 %IdlingKw=0 %EffCharge=95 %EffDischarge=95 DispMode=External

基準電路建置完成！開始進行 15 分鐘步進模擬 (模式: PSO)...

F12 Response

1.

{status: "success", message: "切換成功", pso_total_cost: 21.94, baseline_total_cost: 2.64,…}

baseline_adjusted_cost: -101.18

baseline_battery_asset_value: 0

baseline_export_kwh: 18.447

baseline_export_revenue: 103.82

baseline_had_overload: false

baseline_total_cost: 2.64

message: "切換成功"

pso_adjusted_cost: -164.5

pso_battery_asset_value: -0.04

pso_export_kwh: 33.135

pso_export_revenue: 186.48

pso_total_cost: 21.94

savings: 63.32

status: "success"

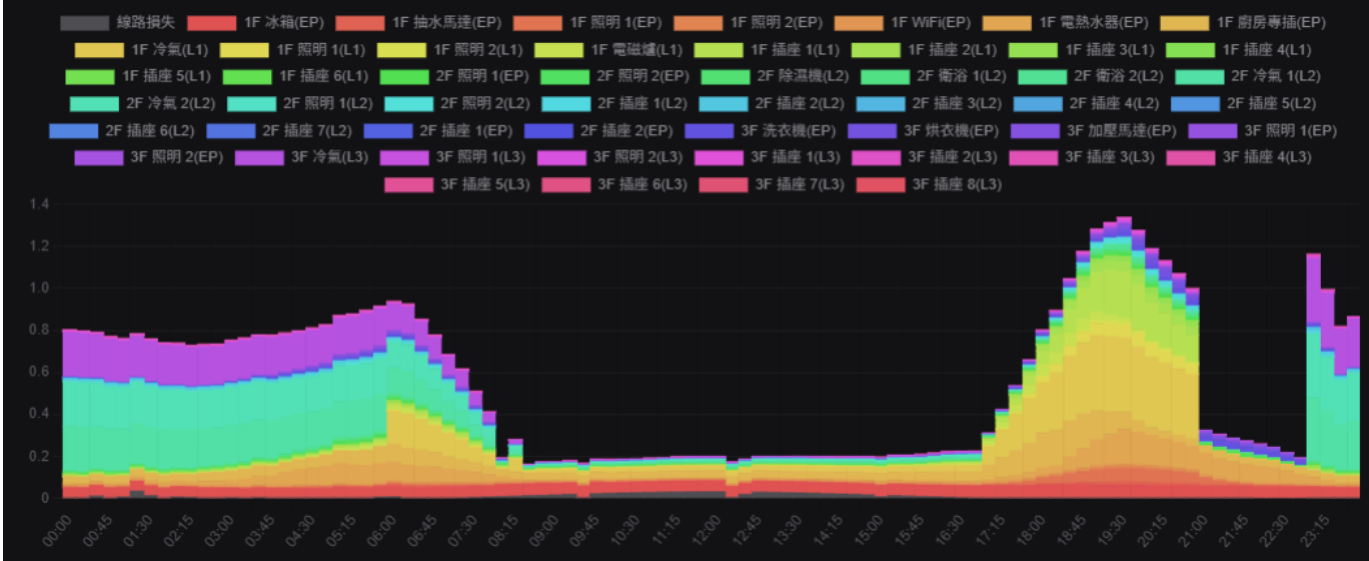
今日總用電量

13.1 度

啟用設備數

35 台

HOURLY LOAD (KW)



優化前電價

NT$ -101.2

優化後電價

NT$ -164.5

省下金額

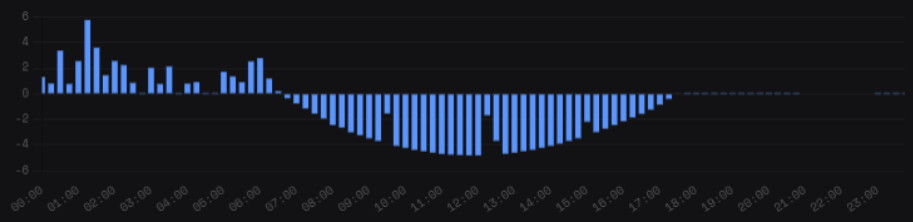
NT$ 63.3

02 — OPTIMIZATION

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1. GRID PURCHASE POWER (KW) - 電網購電

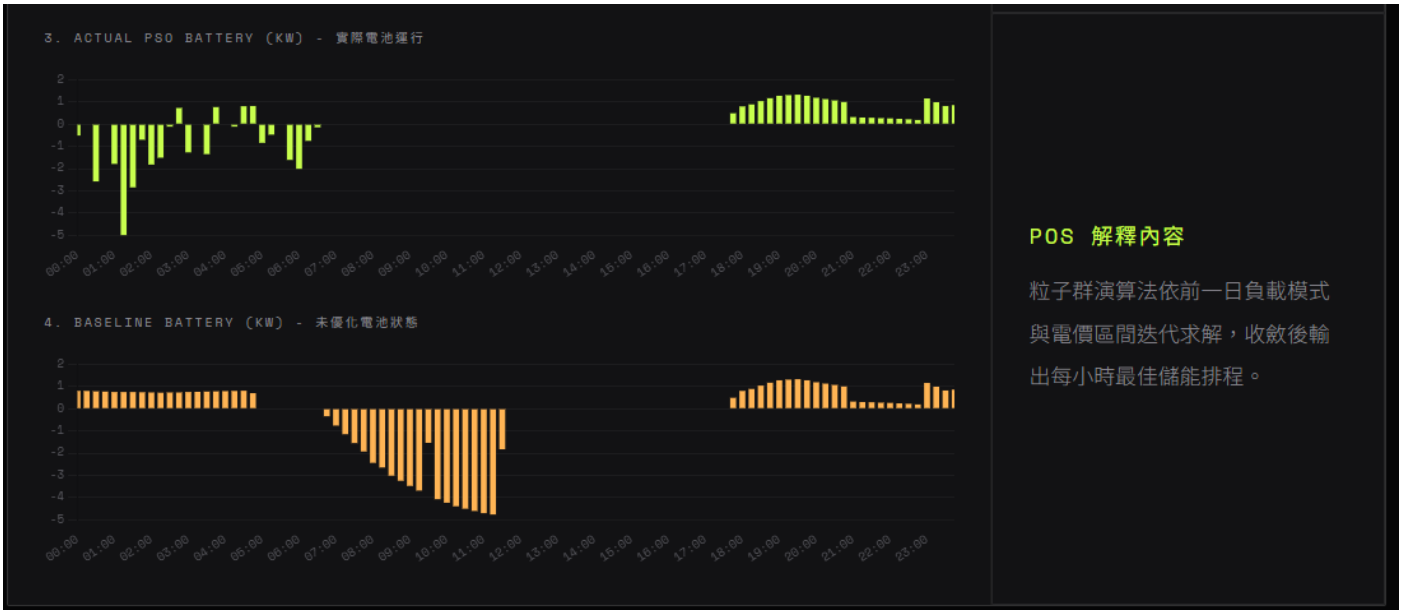


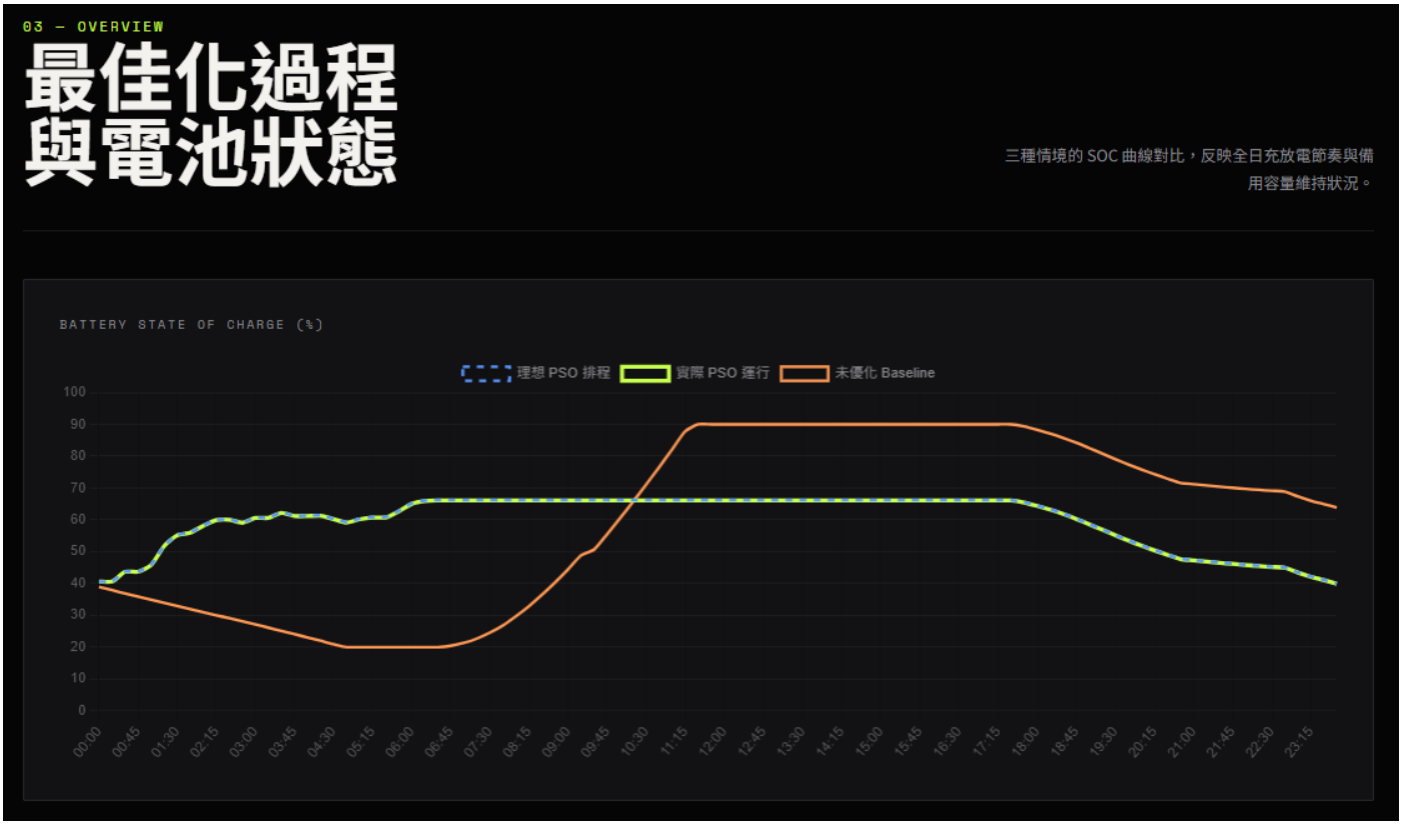
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數據解析

系統逐時段求解電池最佳充放電策略，在維持用電舒適度的前提下最小化尖峰負載與總電費支出。





Pso 的電池紀錄

Time	soc	battery_power_kw	current_a
00:00	40.6	-0.51	2.3
00:15	40.6	0	0
00:30	43.67	-2.58	11.81
00:45	43.67	0	0
01:00	45.79	-1.79	8.19
01:15	51.73	-5	22.93
01:30	55.11	-2.85	13.03
01:45	55.95	-0.7	3.22
02:00	58.11	-1.82	8.32

02:15	59.91	-1.51	6.92
02:30	60.05	-0.11	0.52
02:45	59.08	0.73	3.33
03:00	60.59	-1.27	5.81
03:15	60.59	0	0
03:30	62.2	-1.35	6.17
03:45	61.18	0.77	3.53
04:00	61.18	0	0
04:15	61.3	-0.11	0.49
04:30	60.24	0.81	3.69
04:45	59.15	0.82	3.75
05:00	60.16	-0.84	3.85
05:15	60.72	-0.47	2.15
05:30	60.72	0	0
05:45	62.63	-1.61	7.37
06:00	65.02	-2.01	9.19
06:15	65.91	-0.75	3.41
06:30	66.07	-0.14	0.63
06:45	66.07	0	0
07:00	66.07	0	0
07:15	66.07	0	0
07:30	66.07	0	0
07:45	66.07	0	0
08:00	66.07	0	0
08:15	66.07	0	0
08:30	66.07	0	0
08:45	66.07	0	0
09:00	66.07	0	0
09:15	66.07	0	0
09:30	66.07	0	0
09:45	66.07	0	0
10:00	66.07	0	0
10:15	66.07	0	0
10:30	66.07	0	0
10:45	66.07	0	0
11:00	66.07	0	0
11:15	66.07	0	0
11:30	66.07	0	0
11:45	66.07	0	0
12:00	66.07	0	0
12:15	66.07	0	0
12:30	66.07	0	0

12:45	66.07	0	0
13:00	66.07	0	0
13:15	66.07	0	0
13:30	66.07	0	0
13:45	66.07	0	0
14:00	66.07	0	0
14:15	66.07	0	0
14:30	66.07	0	0
14:45	66.07	0	0
15:00	66.07	0	0
15:15	66.07	0	0
15:30	66.07	0	0
15:45	66.07	0	0
16:00	66.07	0	0
16:15	66.07	0	0
16:30	66.07	0	0
16:45	66.07	0	0
17:00	66.07	0	0
17:15	66.07	0	0
17:30	66.07	0	0
17:45	65.43	0.49	2.21
18:00	64.38	0.8	3.64
18:15	63.21	0.89	4.06
18:30	61.83	1.04	4.76
18:45	60.29	1.17	5.34
19:00	58.61	1.28	5.83
19:15	56.88	1.31	5.97
19:30	55.12	1.33	6.08
19:45	53.45	1.28	5.81
20:00	51.88	1.19	5.41
20:15	50.4	1.13	5.14
20:30	48.99	1.07	4.86
20:45	47.68	1	4.54
21:00	47.26	0.32	1.46
21:15	46.86	0.3	1.37
21:30	46.49	0.29	1.3
21:45	46.13	0.27	1.23
22:00	45.79	0.26	1.17
22:15	45.48	0.24	1.1
22:30	45.19	0.22	0.98
22:45	44.94	0.19	0.87
23:00	43.42	1.16	5.28

23:15	42.11	0.99	4.51
23:30	41.04	0.82	3.72
23:45	39.9	0.86	3.94

Baseline 的電池紀錄

Time	soc	battery_power_kw	current_a
00:00	38.95	0.8	3.65
00:15	37.9	0.8	3.63
00:30	36.87	0.78	3.55
00:45	35.86	0.77	3.5
01:00	34.87	0.75	3.44
01:15	33.88	0.75	3.41
01:30	32.9	0.75	3.4
01:45	31.93	0.74	3.36
02:00	30.97	0.73	3.33
02:15	30.02	0.72	3.29
02:30	29.06	0.73	3.33
02:45	28.09	0.73	3.33
03:00	27.11	0.75	3.41
03:15	26.11	0.76	3.47
03:30	25.09	0.77	3.53
03:45	24.07	0.78	3.53
04:00	23.04	0.79	3.58
04:15	21.99	0.8	3.63
04:30	20.92	0.81	3.69
04:45	20	0.7	3.19
05:00	20	0	0
05:15	20	0	0
05:30	20	0	0
05:45	20	0	0
06:00	20	0	0
06:15	20	0	0
06:30	20	0	0
06:45	20.41	-0.34	1.56
07:00	21.3	-0.76	3.44
07:15	22.67	-1.15	5.24
07:30	24.51	-1.55	7.07
07:45	26.8	-1.93	8.79
08:00	29.7	-2.44	11.14
08:15	32.85	-2.65	12.06
08:30	36.43	-3.02	13.75

08:45	40.28	-3.25	14.79
09:00	44.42	-3.48	15.86
09:15	48.81	-3.69	16.84
09:30	50.64	-1.54	7.03
09:45	55.48	-4.08	18.59
10:00	60.52	-4.24	19.34
10:15	65.74	-4.4	20.04
10:30	71.1	-4.51	20.55
10:45	76.57	-4.61	21.02
11:00	82.17	-4.71	21.48
11:15	87.83	-4.77	21.73
11:30	90	-1.83	8.29
11:45	90	0	0
12:00	90	0	0
12:15	90	0	0
12:30	90	0	0
12:45	90	0	0
13:00	90	0	0
13:15	90	0	0
13:30	90	0	0
13:45	90	0	0
14:00	90	0	0
14:15	90	0	0
14:30	90	0	0
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15:15	90	0	0
15:30	90	0	0
15:45	90	0	0
16:00	90	0	0
16:15	90	0	0
16:30	90	0	0
16:45	90	0	0
17:00	90	0	0
17:15	90	0	0
17:30	90	0	0
17:45	89.36	0.49	2.21
18:00	88.31	0.8	3.64
18:15	87.14	0.89	4.06
18:30	85.76	1.04	4.76
18:45	84.22	1.17	5.34
19:00	82.53	1.28	5.83

19:15	80.81	1.31	5.97
19:30	79.05	1.33	6.08
19:45	77.37	1.28	5.81
20:00	75.81	1.19	5.41
20:15	74.33	1.13	5.14
20:30	72.92	1.07	4.86
20:45	71.61	1	4.54
21:00	71.19	0.32	1.46
21:15	70.79	0.3	1.37
21:30	70.41	0.29	1.3
21:45	70.06	0.27	1.23
22:00	69.72	0.26	1.17
22:15	69.4	0.24	1.1
22:30	69.12	0.22	0.98
22:45	68.87	0.19	0.87
23:00	67.34	1.16	5.28
23:15	66.04	0.99	4.51
23:30	64.96	0.82	3.72
23:45	63.83	0.86	3.94

S7 測試 2 段式 無累計電價 無停電 有設備啟用 冷氣 14:00-16:00

📩 收到的 config： {'pricing': 'two_stage', 'season': 'summer', 'day_type': 'weekday', 'weather': 'sunny', 'monthly_cumulative_kwh': 0, 'outage': None, 'device_schedules': [{'column': 'l2_airc2_abn', 'start_time': '14:00', 'end_time': '16:00'}]}

🔁 [重試檢查] 第 0 次嘗試，目前 bess_override=None

🔎 [天氣檢查] PSO 這次使用的 PV 檔案：C:\projects\hems-simulation-web\data\01_raw\pv_curve_15min.csv

🔎 [PSO 輸入比對] 13:00~15:00 PV: [4.78867, 4.6735299999999995, 4.56147, 4.415, 4.25413, 4.07467, 3.8710700000000005, 3.6598]

🔎 [PSO 輸入比對] 13:00~15:00 critical_load_kw: [0.10398015000000001, 0.104320565, 0.105352634, 0.10483838199999998, 0.105803373, 0.10691419400000002, 0.10865781599999999, 0.11004641800000001]

🎯 [PSO 自評] PSO 演算法自己算出來的理論總成本(電費+懲罰項): -149.19849840594375

🔎 [天氣檢查] 離線驗證這次使用的 PV 檔案：pv_curve_15min.csv

🔍 [除錯] 目前實際使用的電池指令是: New Storage.Battery_Sys phases=1 bus1=Inv_AC.1.2 kV=0.22 kVA=6.0 kWrated=5.0 kWhrated=20.0 pf=1.0 %stored=40 %reserve=0 %IdlingKw=0 %EffCharge=95 %EffDischarge=95 DispMode=External

基準電路建置完成！開始進行 15 分鐘步進模擬 (模式: PSO)...

F12 Response

1. {status: "success", message: "切換成功", pso_total_cost: 24.7, baseline_total_cost: 2.65,…}
   - baseline_adjusted_cost: -87.83
   - baseline_battery_asset_value: 0
   - baseline_export_kwh: 16.077
   - baseline_export_revenue: 90.48
   - baseline_had_overload: false
   - baseline_total_cost: 2.65
   - message: "切換成功"
   - pso_adjusted_cost: -148.45
   - pso_battery_asset_value: 0
   - pso_export_kwh: 30.766
   - pso_export_revenue: 173.15
   - pso_total_cost: 24.7
   - savings: 60.62
   - status: "success"

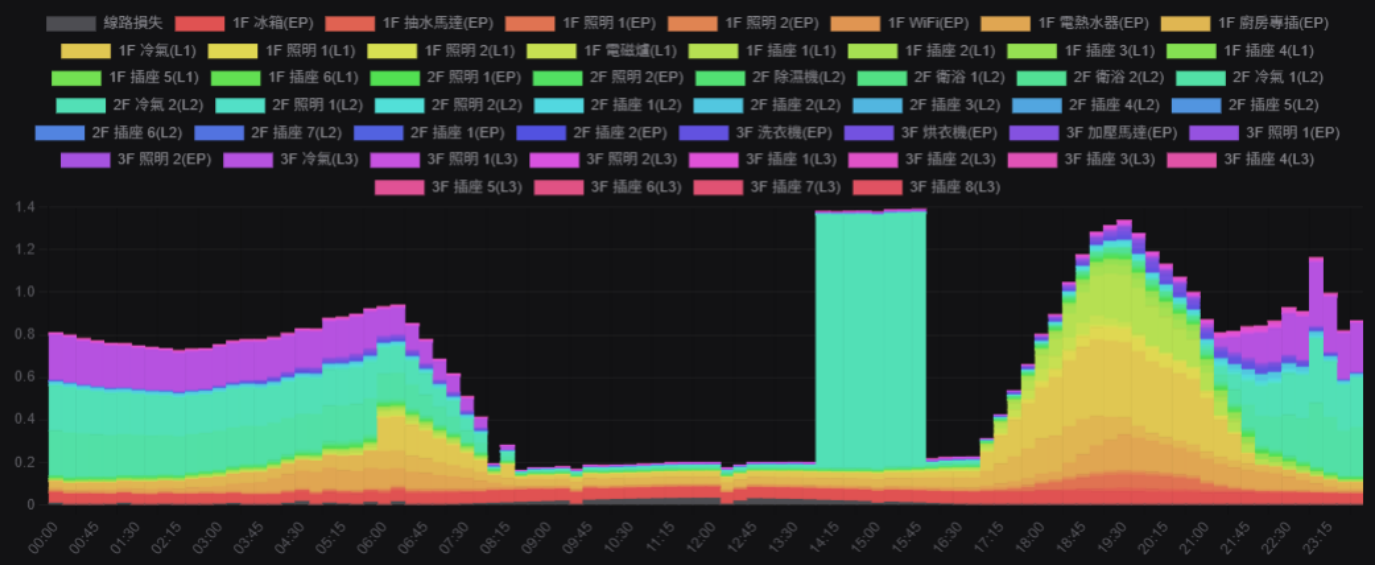
今日總用電量

16.6 度

啟用設備數

35 台

HOURLY LOAD (KW)



優化前電價

NT$ -87.8

優化後電價

NT$ -148.4

省下金額

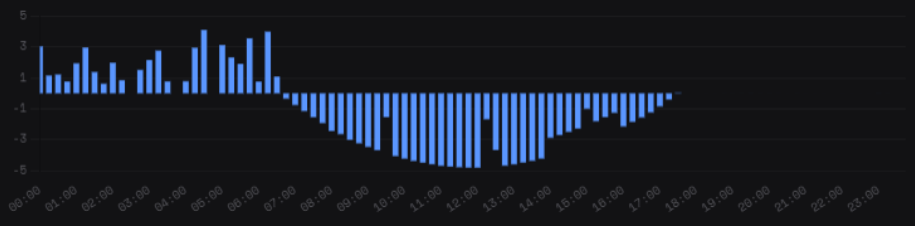
NT$ 60.6

02 — OPTIMIZATION

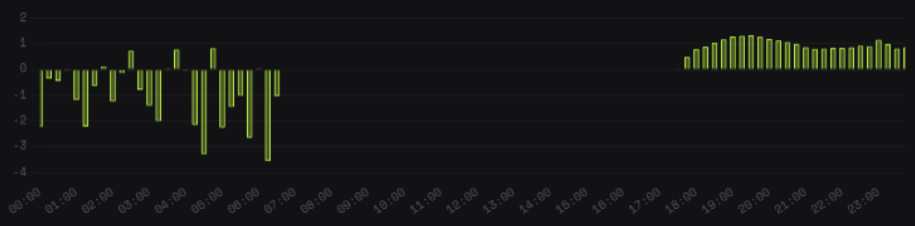
POS 儲能排程最佳化

於用電低谷引導放電、太陽能充裕時段安排充電，降低尖峰購電功率並平滑全日負載。

1. GRID PURCHASE POWER (KW) - 電網購電

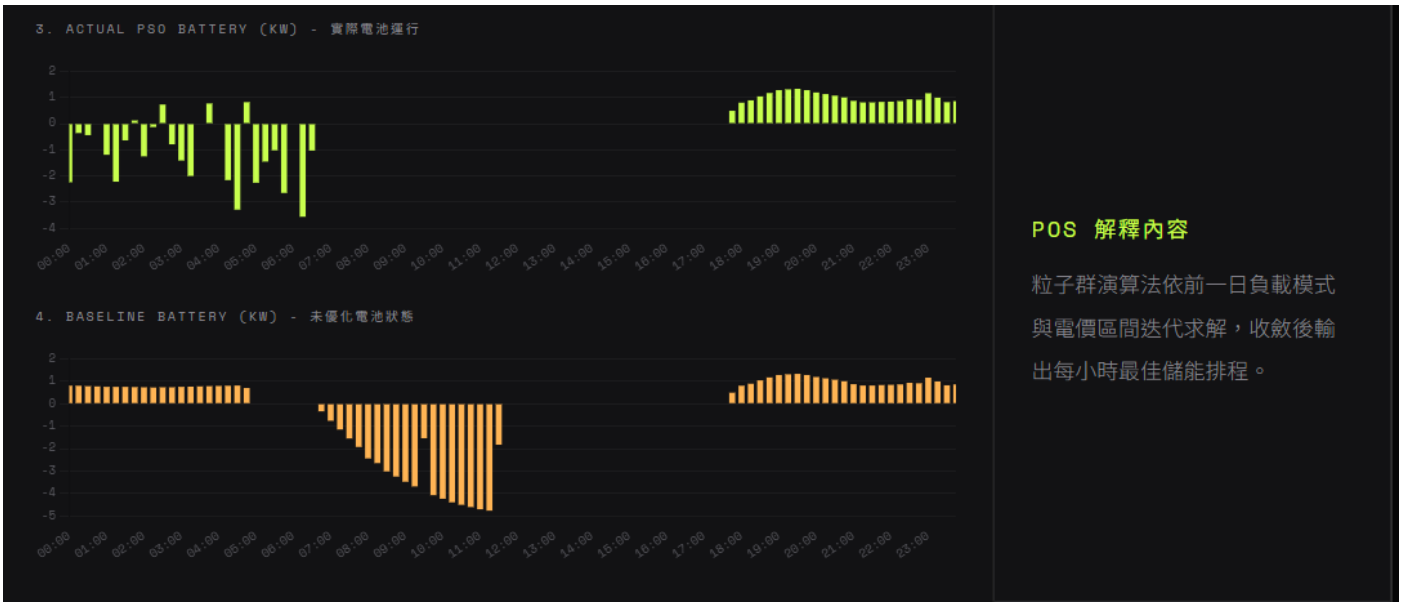


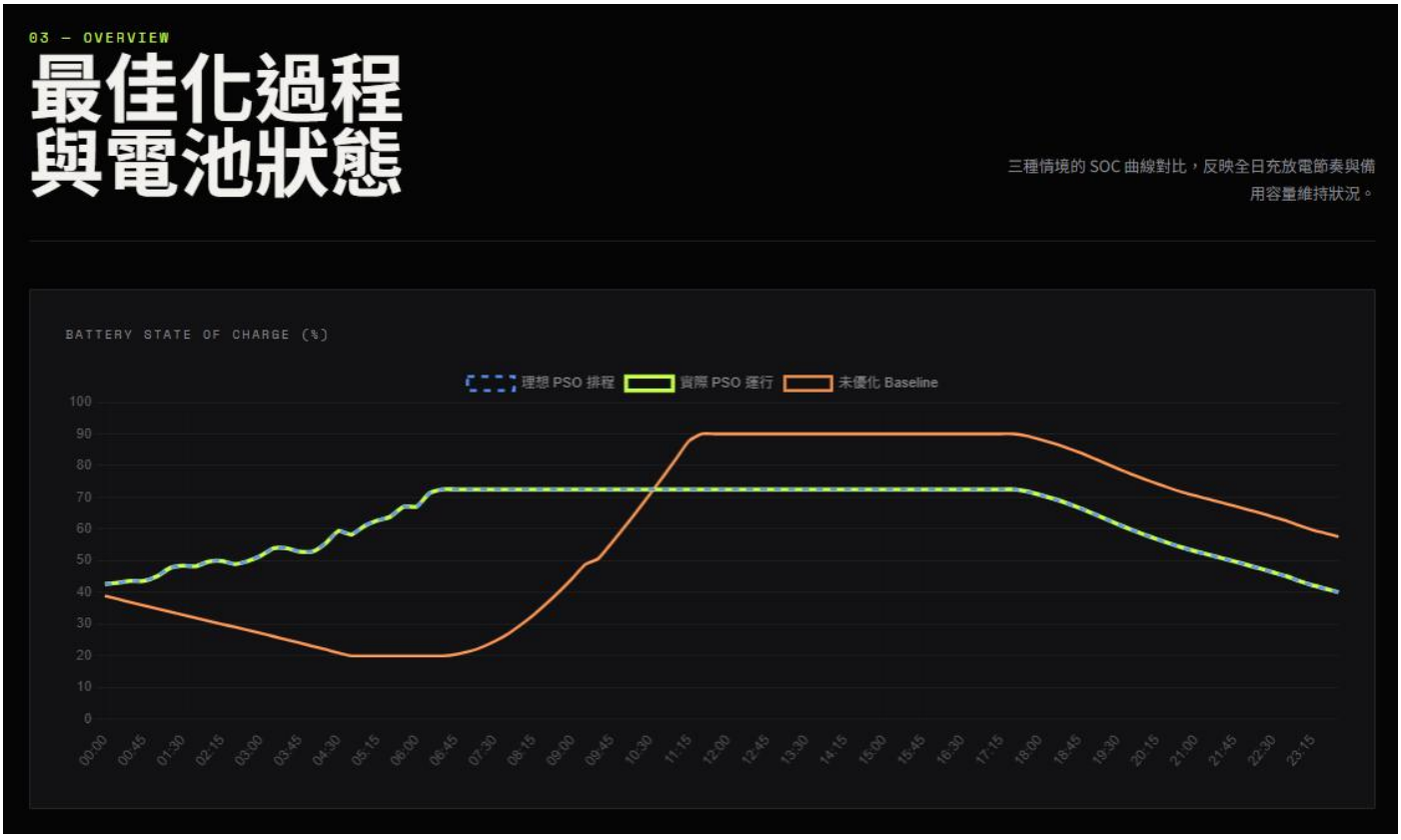
2. IDEAL PSO BATTERY (KW) - 理想電池排程



數據解析

系統逐時段求解電池最佳充放電策略，在維持用電舒適度的前提下最小化尖峰負載與總電費支出。





pso 起始與結束電量

Time	soc	battery_power_kw	current_a
00:00	42.65	-2.24	10.22
00:15	43.08	-0.36	1.63
00:30	43.6	-0.44	2.01
00:45	43.6	0	0
01:00	45.02	-1.19	5.43
01:15	47.65	-2.22	10.14
01:30	48.41	-0.64	2.9
01:45	48.25	0.12	0.53
02:00	49.74	-1.25	5.71

02:15	49.89	-0.13	0.58
02:30	48.93	0.73	3.33
02:45	49.86	-0.79	3.59
03:00	51.53	-1.41	6.44
03:15	53.91	-2	9.16
03:30	53.91	0	0
03:45	52.89	0.77	3.53
04:00	52.89	0	0
04:15	55.45	-2.16	9.85
04:30	59.36	-3.29	15.07
04:45	58.28	0.82	3.75
05:00	60.96	-2.26	10.33
05:15	62.69	-1.45	6.64
05:30	63.9	-1.02	4.64
05:45	67.05	-2.65	12.14
06:00	67.05	0	0
06:15	71.28	-3.56	16.3
06:30	72.5	-1.03	4.71
06:45	72.5	0	0
07:00	72.5	0	0
07:15	72.5	0	0
07:30	72.5	0	0
07:45	72.5	0	0
08:00	72.5	0	0
08:15	72.5	0	0
08:30	72.5	0	0
08:45	72.5	0	0
09:00	72.5	0	0
09:15	72.5	0	0
09:30	72.5	0	0
09:45	72.5	0	0
10:00	72.5	0	0
10:15	72.5	0	0
10:30	72.5	0	0
10:45	72.5	0	0
11:00	72.5	0	0
11:15	72.5	0	0
11:30	72.5	0	0
11:45	72.5	0	0
12:00	72.5	0	0
12:15	72.5	0	0
12:30	72.5	0	0

12:45	72.5	0	0
13:00	72.5	0	0
13:15	72.5	0	0
13:30	72.5	0	0
13:45	72.5	0	0
14:00	72.5	0	0
14:15	72.5	0	0
14:30	72.5	0	0
14:45	72.5	0	0
15:00	72.5	0	0
15:15	72.5	0	0
15:30	72.5	0	0
15:45	72.5	0	0
16:00	72.5	0	0
16:15	72.5	0	0
16:30	72.5	0	0
16:45	72.5	0	0
17:00	72.5	0	0
17:15	72.5	0	0
17:30	72.5	0	0
17:45	71.86	0.49	2.21
18:00	70.81	0.8	3.64
18:15	69.64	0.89	4.06
18:30	68.26	1.04	4.76
18:45	66.72	1.17	5.34
19:00	65.03	1.28	5.83
19:15	63.31	1.31	5.97
19:30	61.55	1.33	6.08
19:45	59.87	1.28	5.81
20:00	58.31	1.19	5.41
20:15	56.83	1.13	5.14
20:30	55.42	1.07	4.86
20:45	54.11	1	4.54
21:00	52.97	0.87	3.95
21:15	51.91	0.81	3.67
21:30	50.84	0.81	3.71
21:45	49.74	0.83	3.8
22:00	48.63	0.84	3.83
22:15	47.5	0.86	3.92
22:30	46.28	0.93	4.22
22:45	45.09	0.91	4.13
23:00	43.56	1.16	5.28

23:15	42.26	0.99	4.51
23:30	41.19	0.82	3.72
23:45	40.05	0.86	3.94

Baseline 的電池記錄

Time	soc	battery_power_kw	current_a
00:00	38.95	0.8	3.65
00:15	37.9	0.8	3.63
00:30	36.87	0.78	3.55
00:45	35.86	0.77	3.5
01:00	34.87	0.75	3.44
01:15	33.88	0.75	3.41
01:30	32.9	0.75	3.4
01:45	31.93	0.74	3.36
02:00	30.97	0.73	3.33
02:15	30.02	0.72	3.29
02:30	29.06	0.73	3.33
02:45	28.09	0.73	3.33
03:00	27.11	0.75	3.41
03:15	26.11	0.76	3.47
03:30	25.09	0.77	3.53
03:45	24.07	0.78	3.53
04:00	23.04	0.79	3.58
04:15	21.99	0.8	3.63
04:30	20.92	0.81	3.69
04:45	20	0.7	3.19
05:00	20	0	0
05:15	20	0	0
05:30	20	0	0
05:45	20	0	0
06:00	20	0	0
06:15	20	0	0
06:30	20	0	0
06:45	20.41	-0.34	1.56
07:00	21.3	-0.76	3.44
07:15	22.67	-1.15	5.24
07:30	24.51	-1.55	7.07
07:45	26.8	-1.93	8.79
08:00	29.7	-2.44	11.14
08:15	32.85	-2.65	12.06
08:30	36.43	-3.02	13.75
08:45	40.28	-3.25	14.79

09:00	44.42	-3.48	15.86
09:15	48.81	-3.69	16.84
09:30	50.64	-1.54	7.03
09:45	55.48	-4.08	18.59
10:00	60.52	-4.24	19.34
10:15	65.74	-4.4	20.04
10:30	71.1	-4.51	20.55
10:45	76.57	-4.61	21.02
11:00	82.17	-4.71	21.48
11:15	87.83	-4.77	21.73
11:30	90	-1.83	8.29
11:45	90	0	0
12:00	90	0	0
12:15	90	0	0
12:30	90	0	0
12:45	90	0	0
13:00	90	0	0
13:15	90	0	0
13:30	90	0	0
13:45	90	0	0
14:00	90	0	0
14:15	90	0	0
14:30	90	0	0
14:45	90	0	0
15:00	90	0	0
15:15	90	0	0
15:30	90	0	0
15:45	90	0	0
16:00	90	0	0
16:15	90	0	0
16:30	90	0	0
16:45	90	0	0
17:00	90	0	0
17:15	90	0	0
17:30	90	0	0
17:45	89.36	0.49	2.21
18:00	88.31	0.8	3.64
18:15	87.14	0.89	4.06
18:30	85.76	1.04	4.76
18:45	84.22	1.17	5.34
19:00	82.53	1.28	5.83
19:15	80.81	1.31	5.97

19:30	79.05	1.33	6.08
19:45	77.37	1.28	5.81
20:00	75.81	1.19	5.41
20:15	74.33	1.13	5.14
20:30	72.92	1.07	4.86
20:45	71.61	1	4.54
21:00	70.47	0.87	3.95
21:15	69.41	0.81	3.67
21:30	68.34	0.81	3.71
21:45	67.24	0.83	3.8
22:00	66.13	0.84	3.83
22:15	65	0.86	3.92
22:30	63.78	0.93	4.22
22:45	62.59	0.91	4.13
23:00	61.06	1.16	5.28
23:15	59.76	0.99	4.51
23:30	58.68	0.82	3.72
23:45	57.55	0.86	3.94